

	1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch		Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch LED Panel Connectors File: led_panel_connectors.kicad_sch Pushbuttons File: pushbuttons.kicad_sch Mechanical File: mechanical.kicad_sch			
B						
C						
D						
	1	2	3	4	5	

Sheet: / File: Audio_Visualiser_DSP.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
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TRY TO TALLY LOAD CURRENTS

The circuit is a +12V power input regulator using the TPS26600RHFT IC. The input is connected to a +12V IN source through a fuse J1 and a diode D1. The input voltage is monitored by TP2. The IC is configured with UVLO (Under Voltage Lockout) and OVP (Over Voltage Protection) thresholds. The output is +12V, regulated by the IC, and is connected to a 100uF capacitor C3. The output voltage is monitored by TP3. The circuit also includes a Telemetry module and a PWR_FLAG output.

UVLO threshold set to 10.8V
OVLO threshold set to 13.2V
MODE set to Circuit Breaker with Auto-Retry
Output Current Limit set to 1.2A

Sheet: /12V Power Input/
File: POS12_power_input.kicad_sch

Title:

Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 2/20

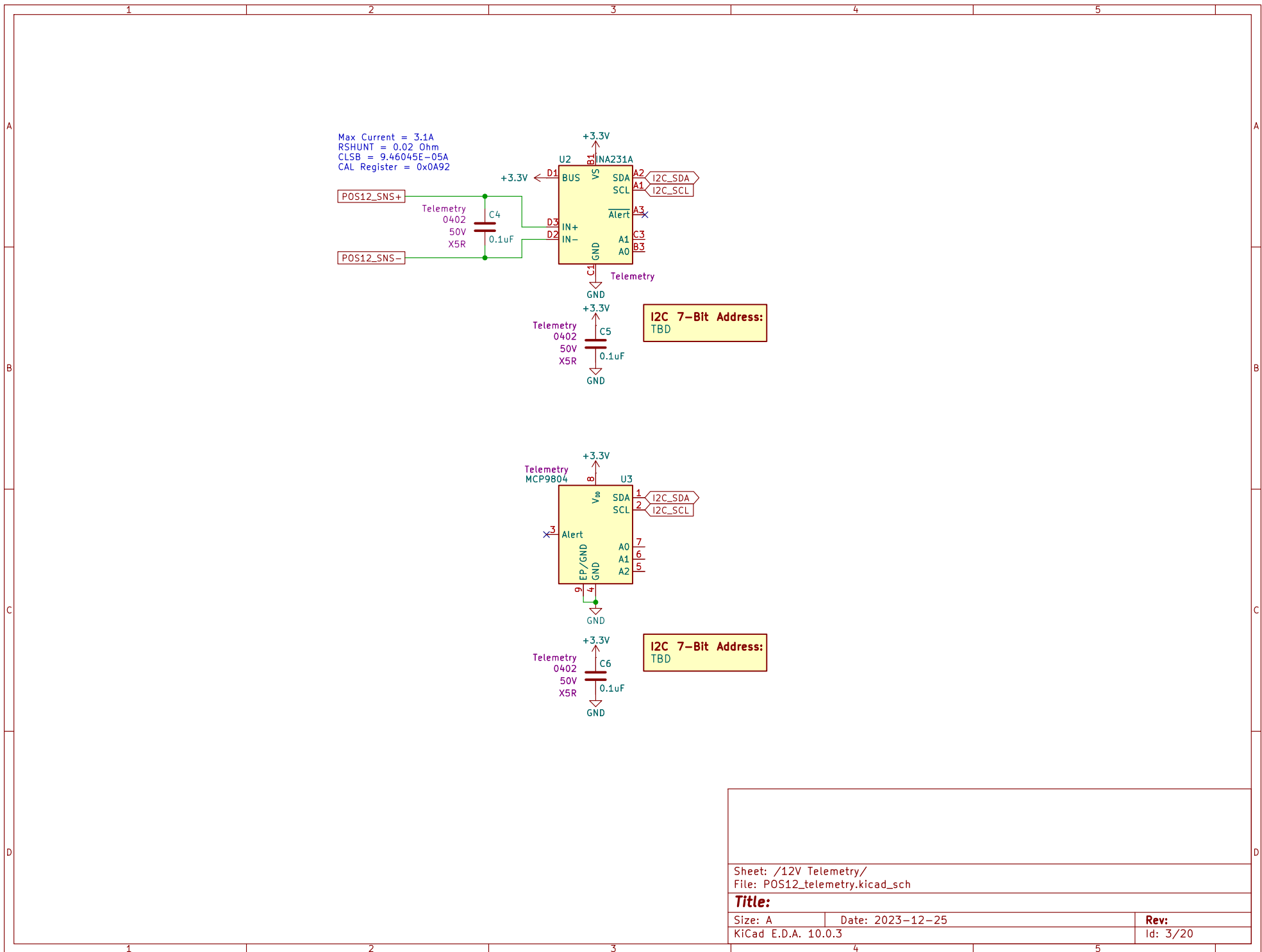
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KICad E.D.A. 10.0.3		Id: 2/20

Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 2/20

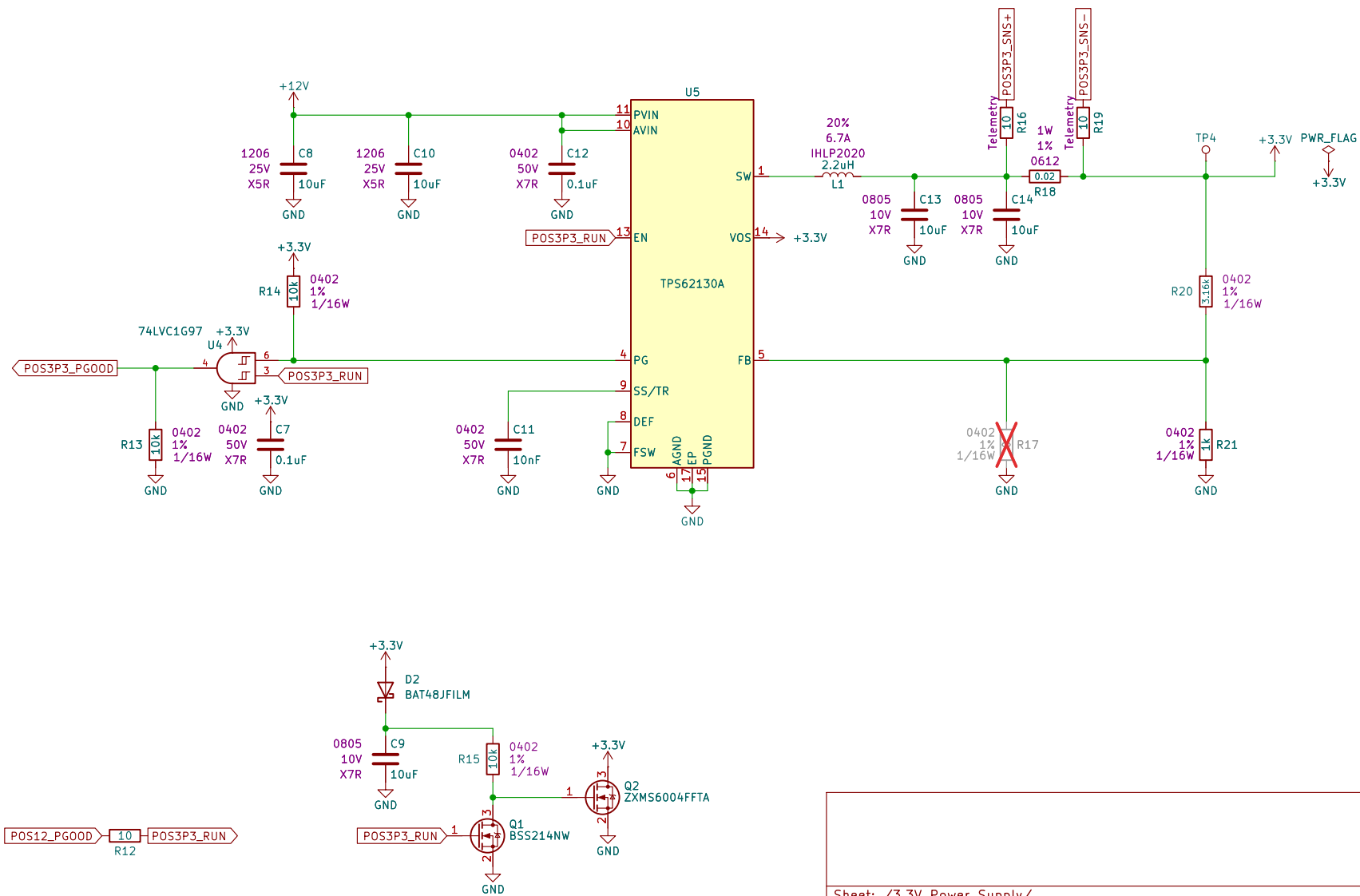
KiCad E.D.A. 10.0.3	Id: 2/20
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[illegible]

Rev:
Id: 2/20



TRY TO TALLY LOAD CURRENTS

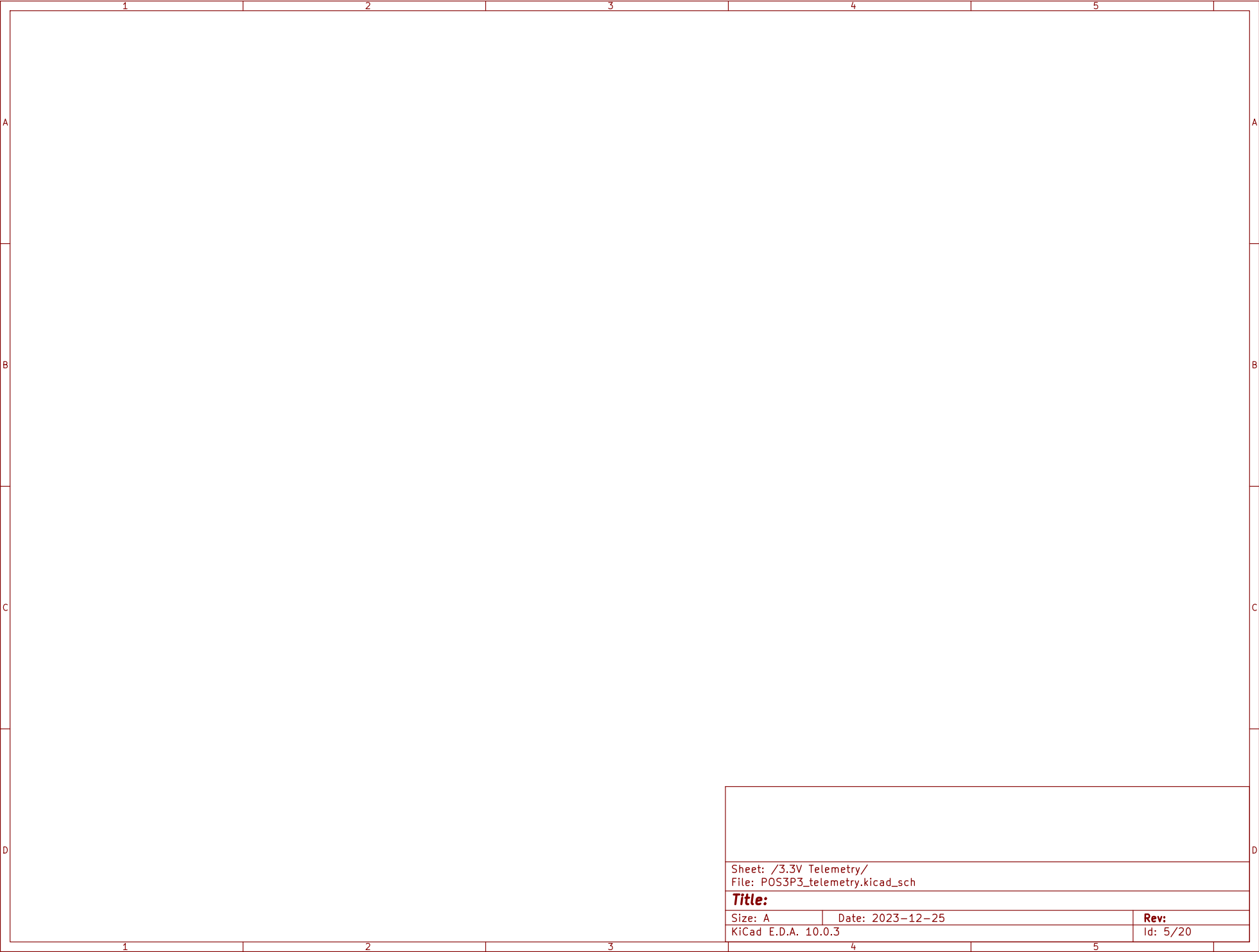


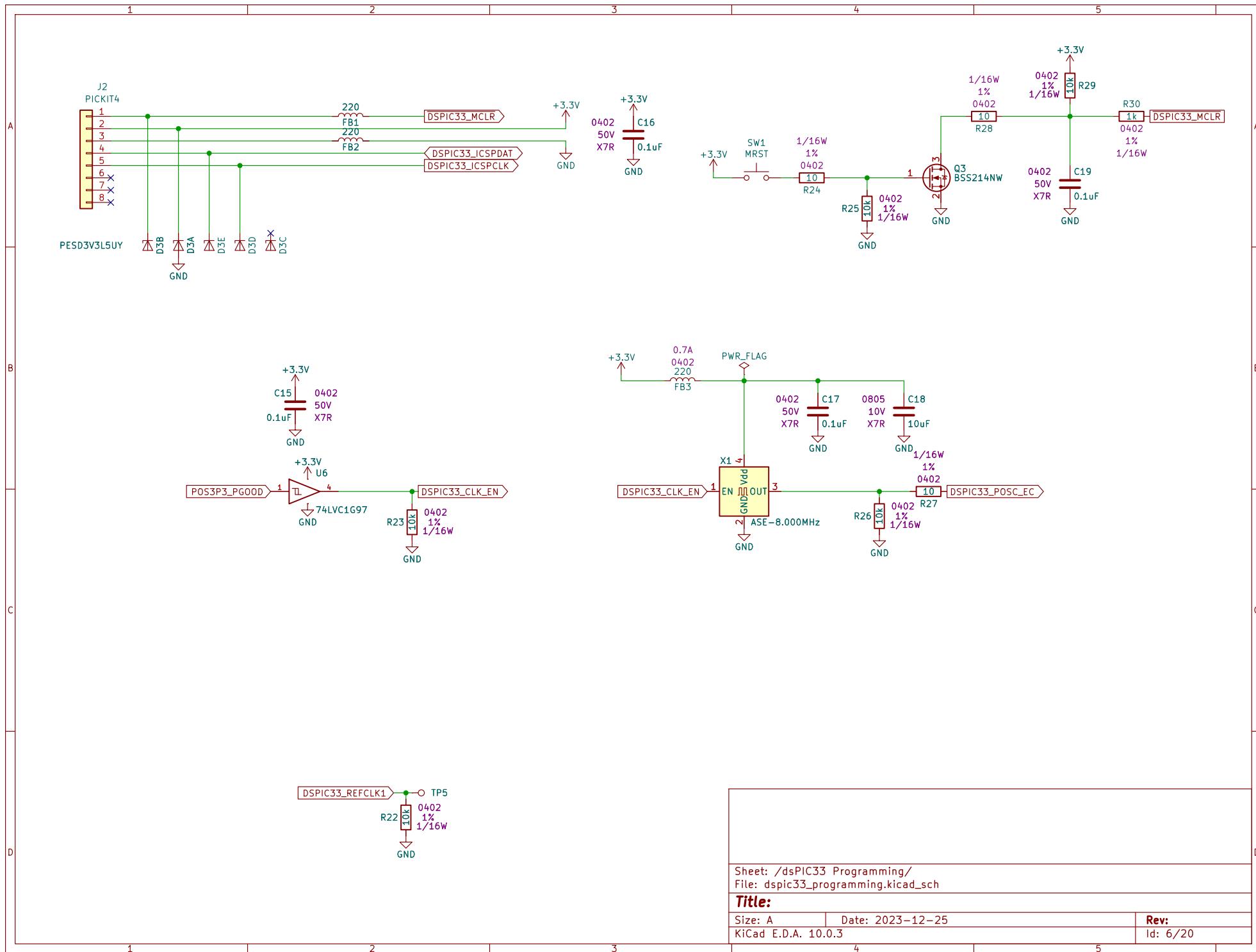
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File: POS3P3_power_supply.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 4/20



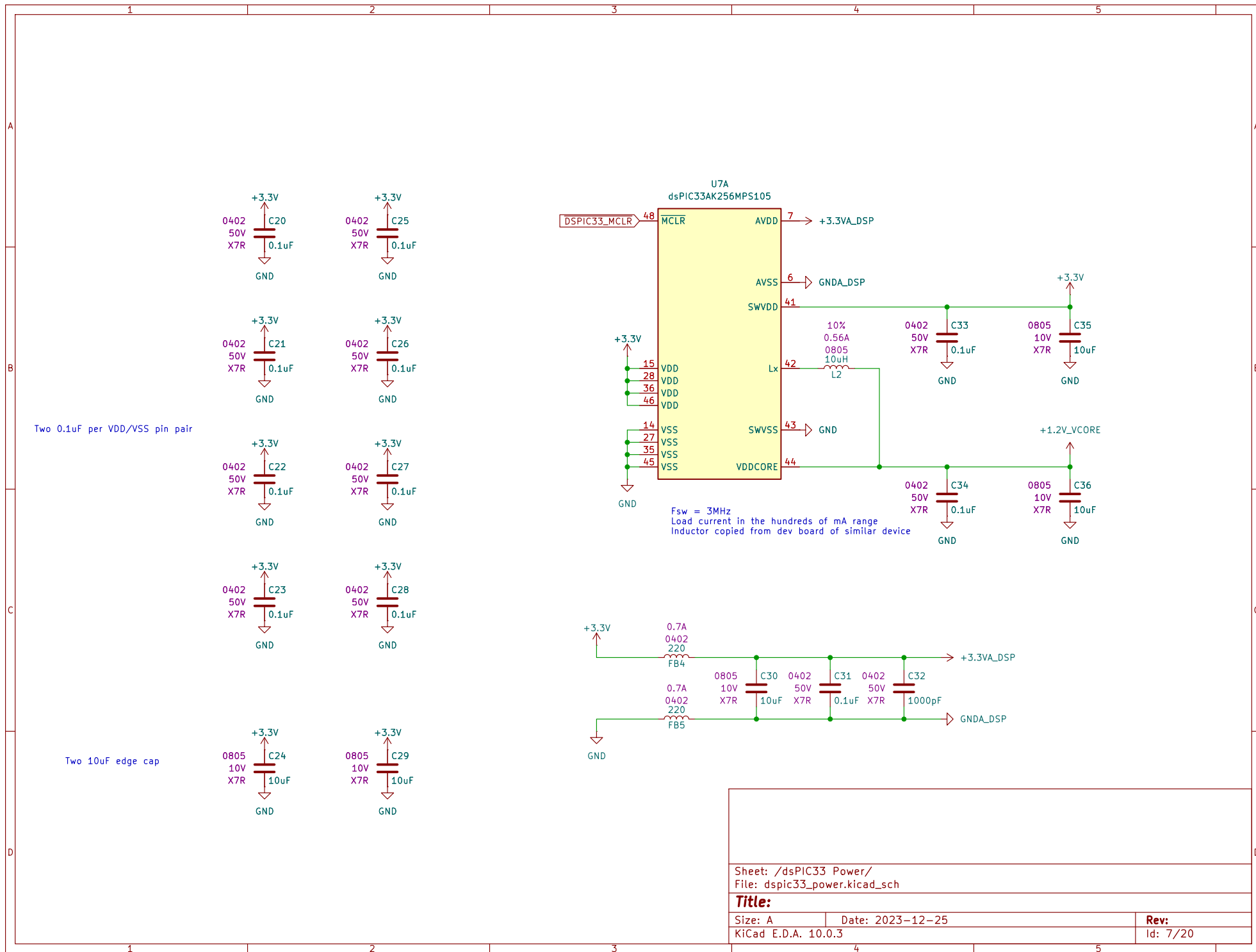


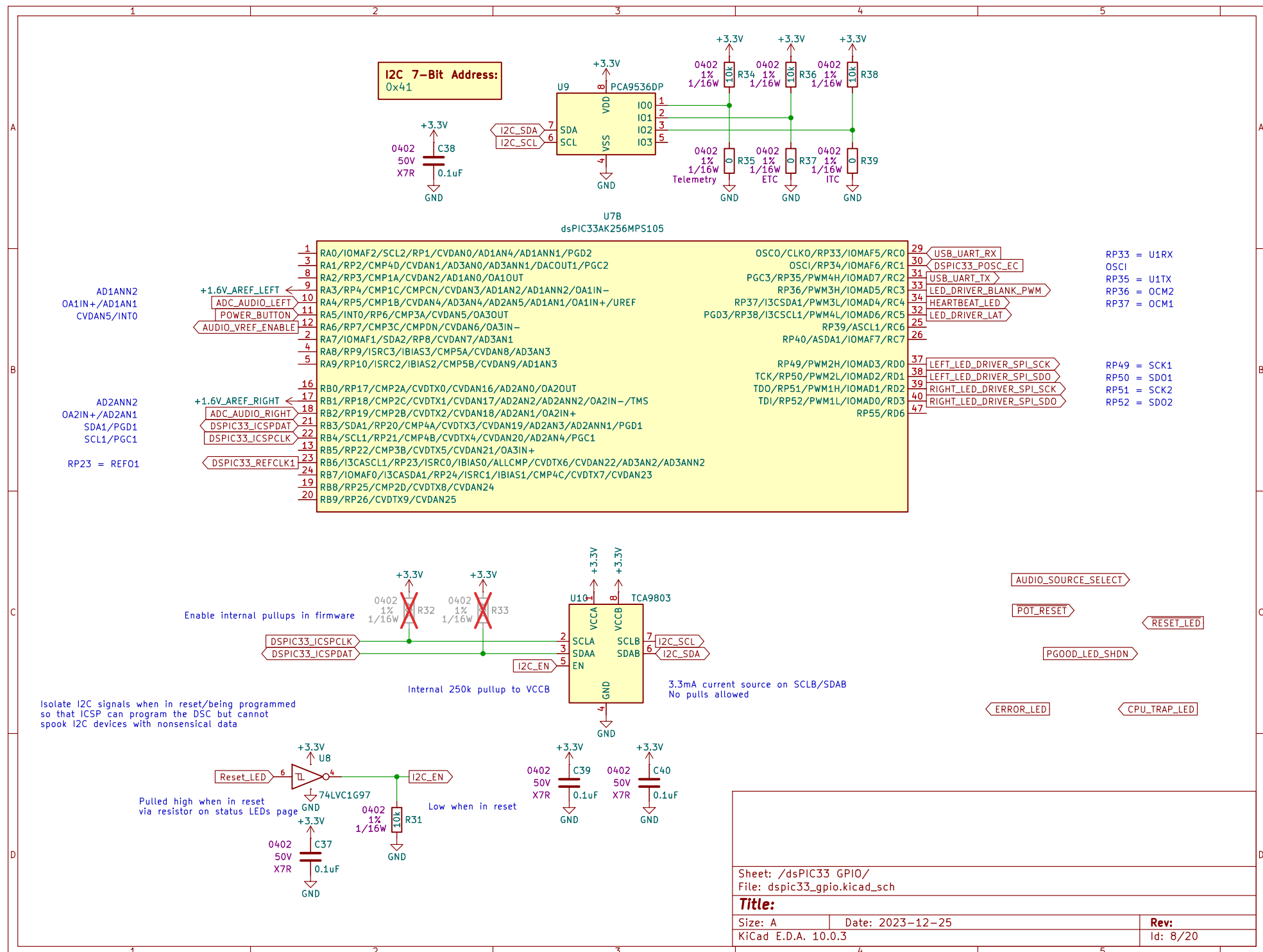
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File: dspic33_programming.kicad_sch

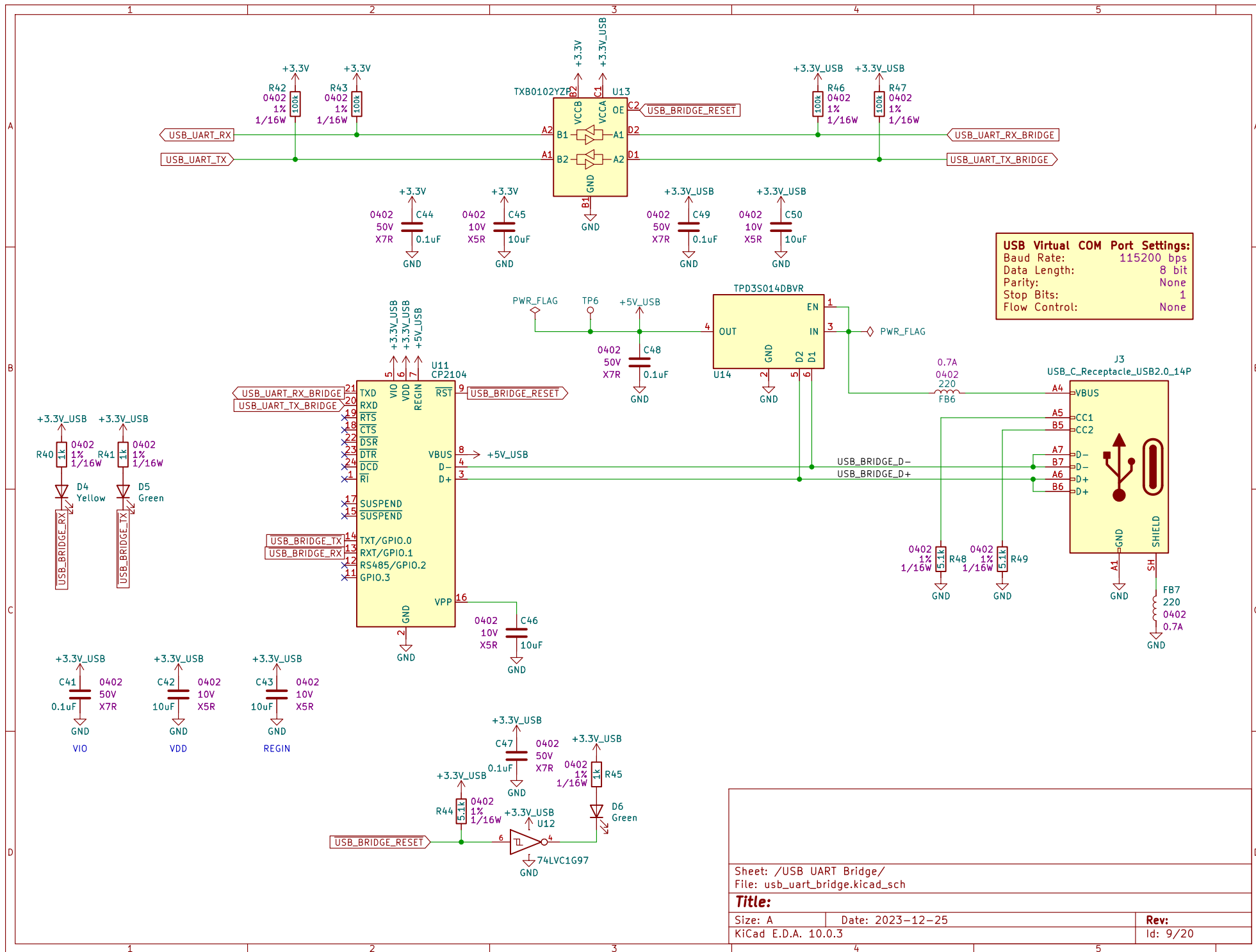
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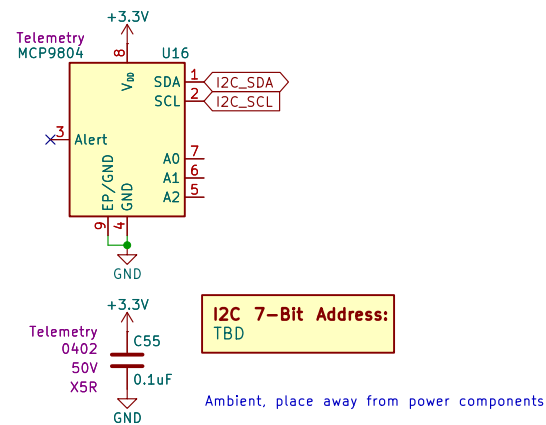
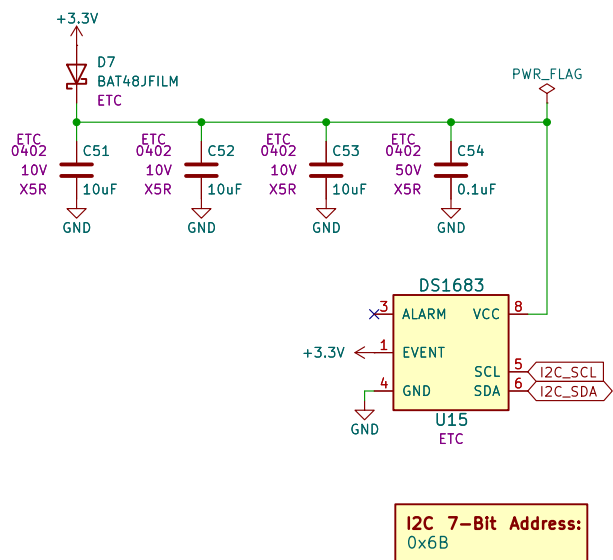
Rev:
Id: 6/20







Sheet: /USB UART Bridge/ File: usb_uart_bridge.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 9/20

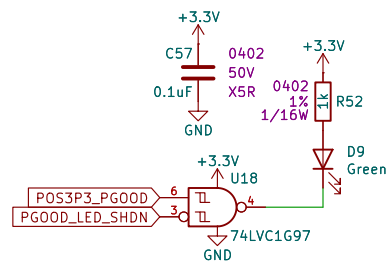
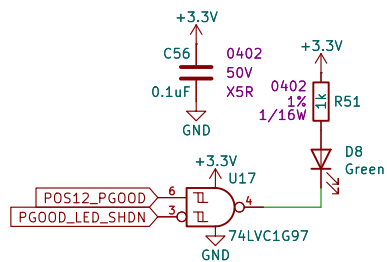
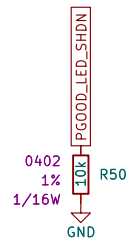


Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 10/20

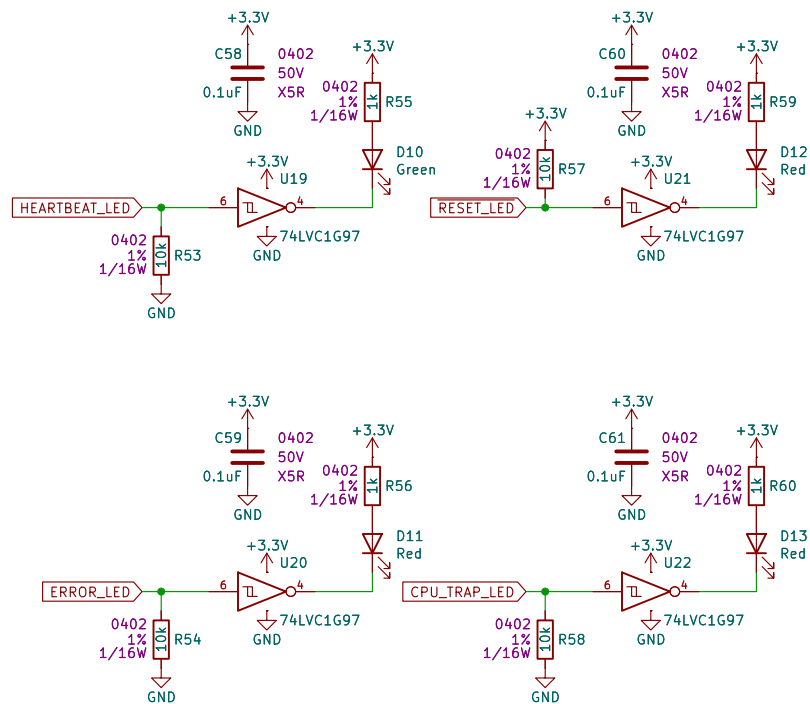


Sheet: /PGOOD LEDs/
File: pgood_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 11/20

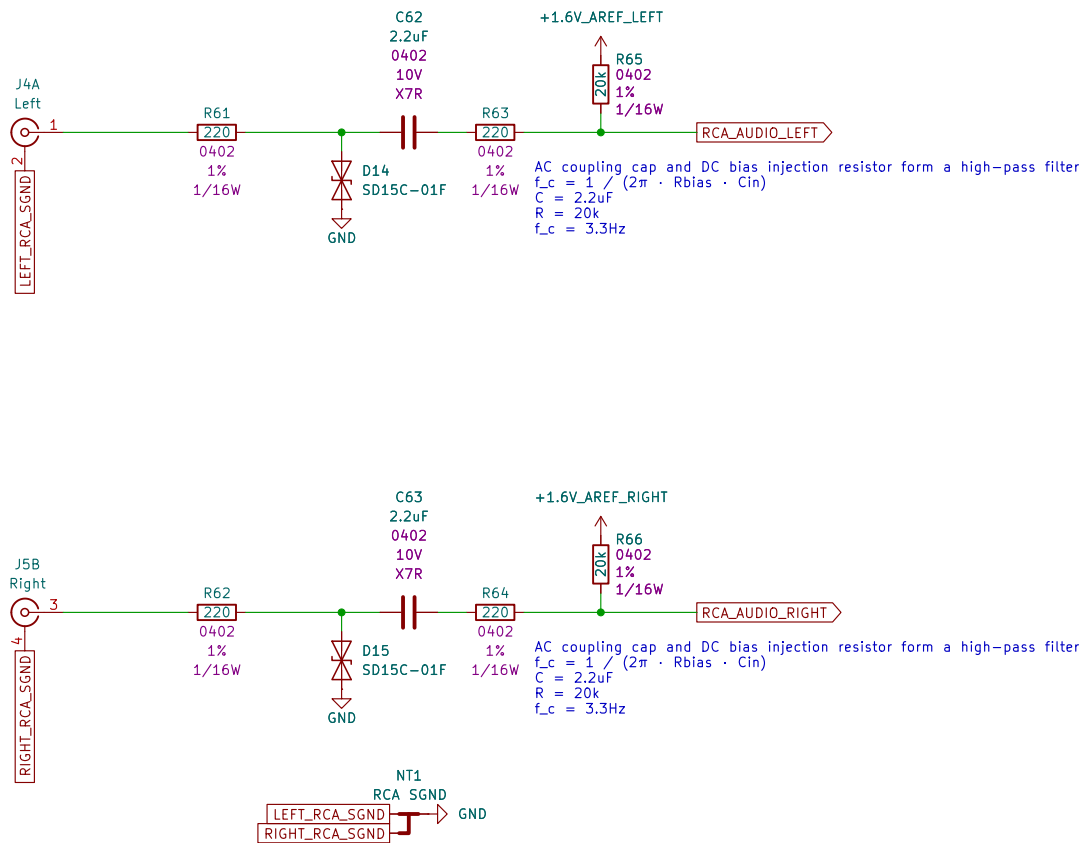


Sheet: /Status LEDs/
File: status_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/20



Sheet: /RCA Signal Conditioning/
File: rca_signal_conditioning.kicad_sch

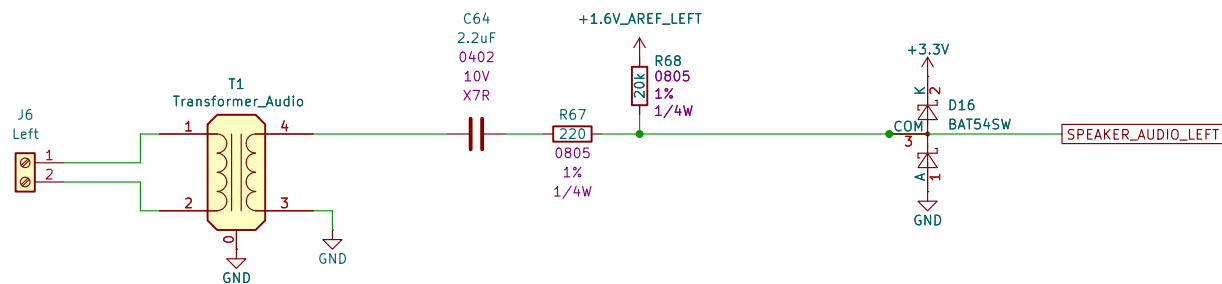
Title:

Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

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Id: 13/20

Alternate is a 20:1 resistive attenuation (for 18V peaks) and a diff amp

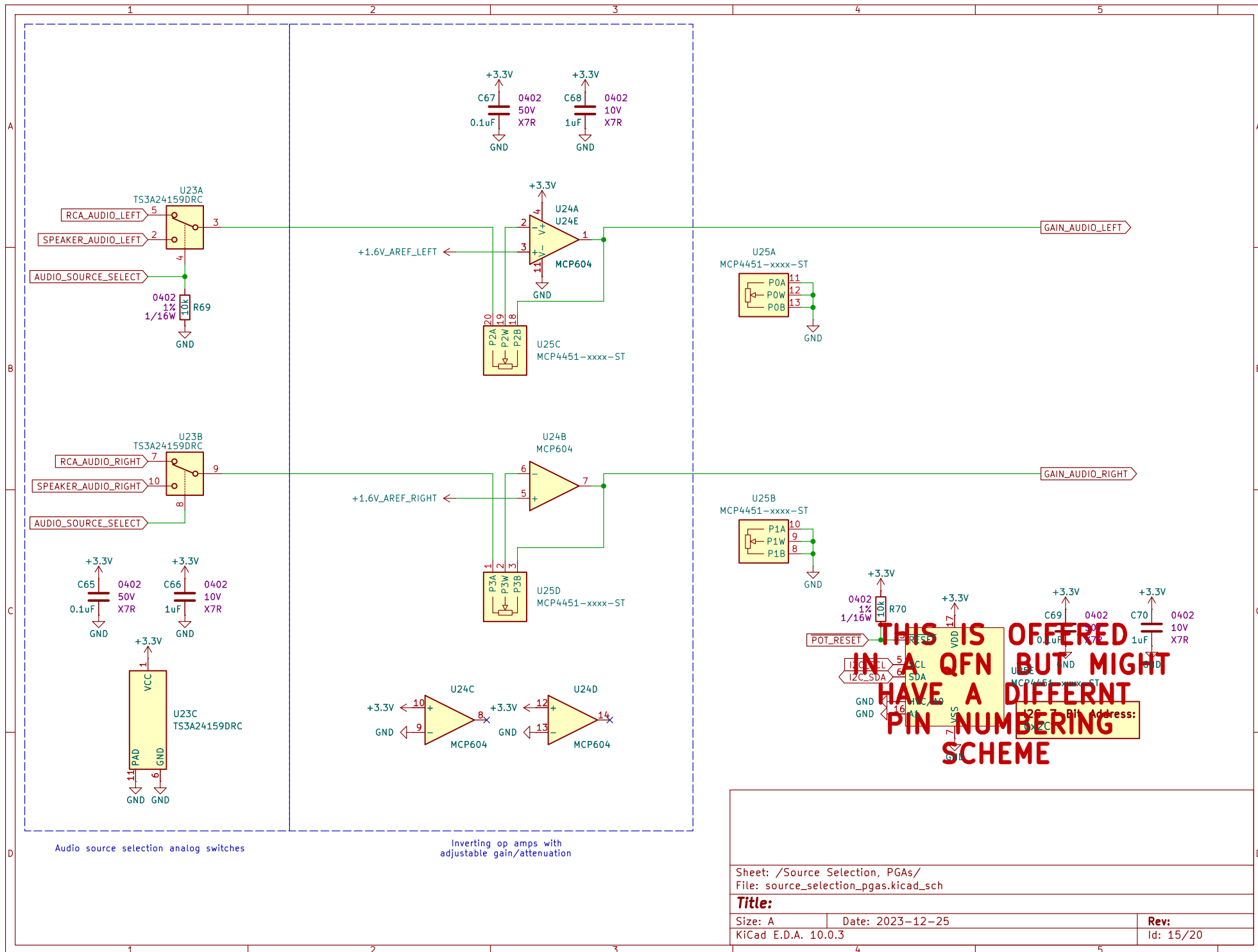


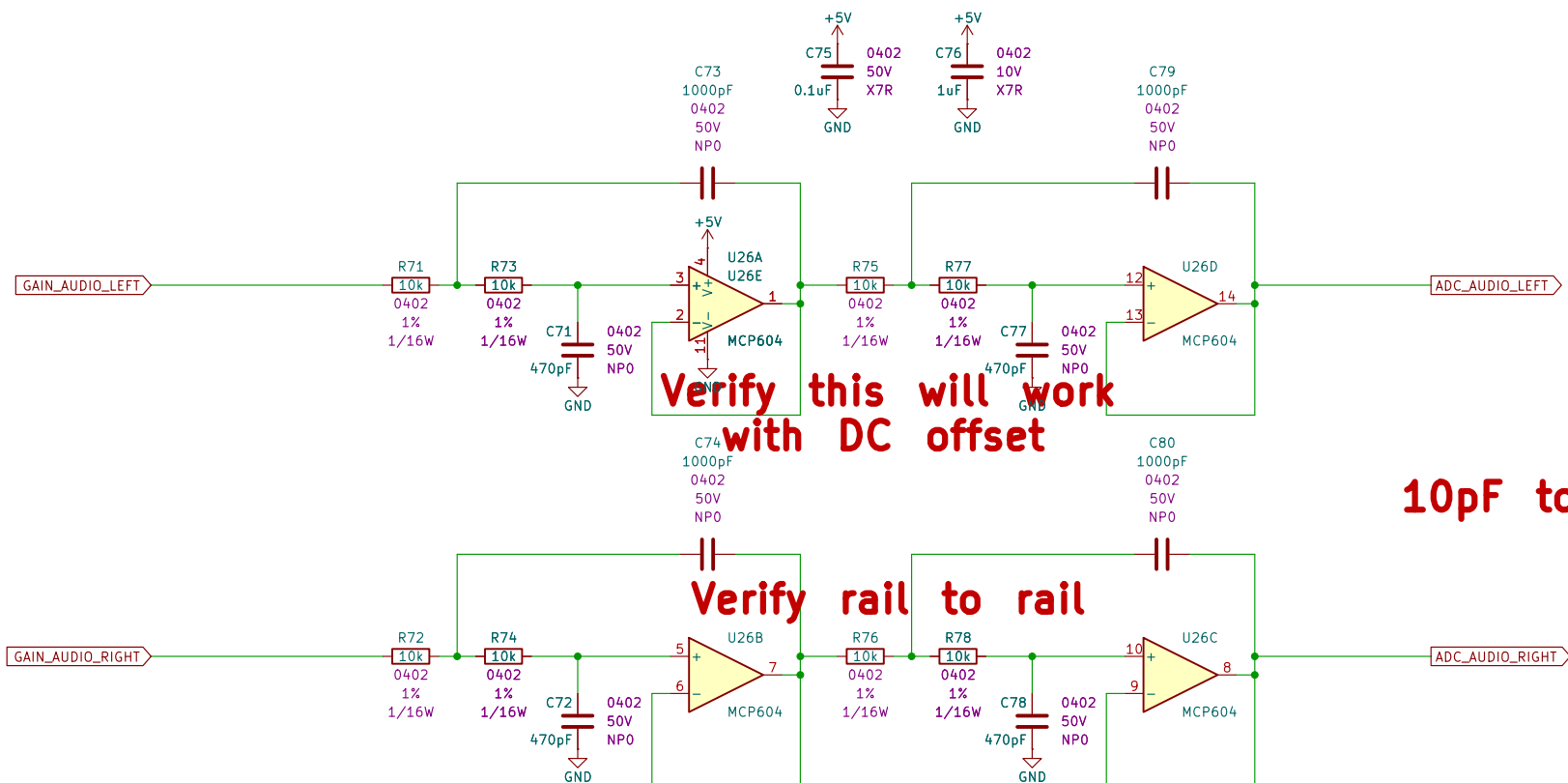
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File: speaker_signal_conditioning.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

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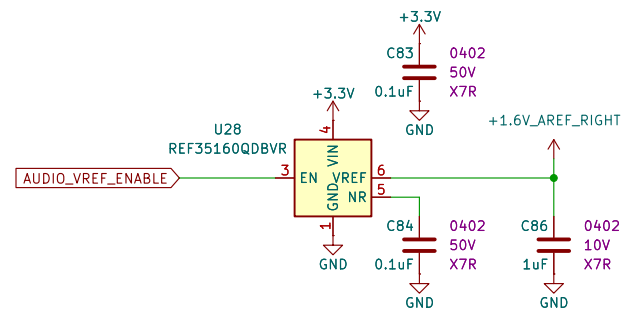
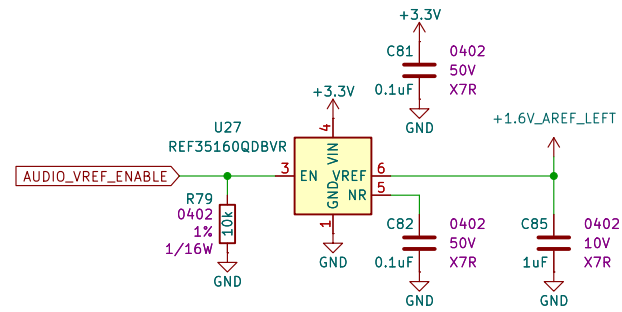
4-pole Sallen-Key active anti-aliasing lowpass filters
 Unity gain
 $f_c = 23.2\text{kHz}$, $Q = 0.707$ each
 If sampling at 44.1kHz , nyquist is 22.05kHz
 Must filter hard just below this

Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 16/20

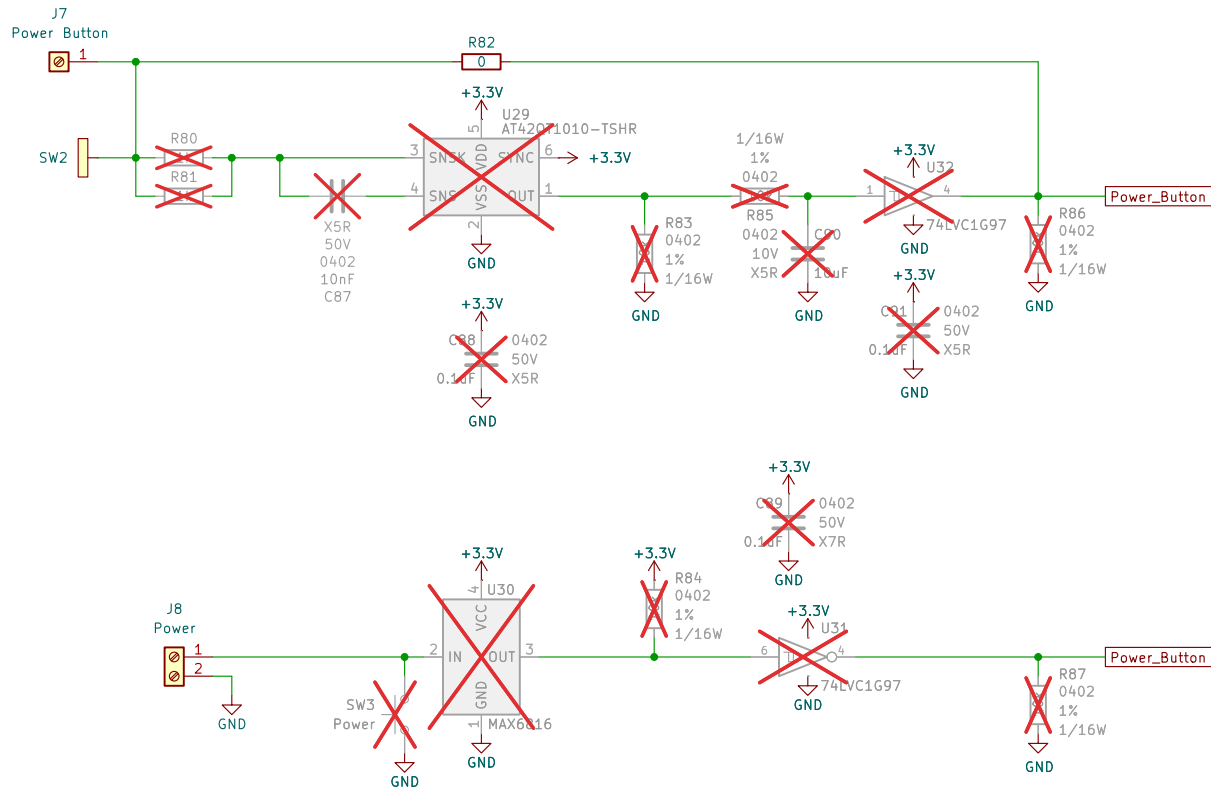


Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 17/20



Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

Title:

Size: A
KiCad E.D.A. 10.0.3

Date:

Rev:

Id: 19/20

